□□□□□

benchmark OOD

frozen representation → 2D scorer → legal structure + confidence

01 评测卫生与主结果 03–04

→ benchmark

02 效率与机制 05–09

→ OOD → ECE → -

03 架构拆解 10

04 迁移与外部验证 11–15

NucleicBERT → RiNALMo → CHANRG →

05 结论与复现 16–23

benchmark

benchmark

OOD

split CHANRG

ECE

NucleicBERT RiNALMo

SOT

0

		TS0	new	TS1	TS2	TS3	hard	TSB	
RNAformer-bprna	ref_tr_bprna (34.9k)	0	0	38/63	28/39	15/19	20/28	339/428	TS1-3/hard/TSB
RNAformer-interfam	ref_tr_inter (2.2k)	0	0	0	0	0	0	1/428	TSB
RiNALMo-ft 4	bprna_tr0 (10.7k)	0	0	0	0	0	0	247/428	TSB
eFold	bpRNA +ArchII	0	0	0	0	0	0	247/428*	TSB
xens2	bprna_tr1c (42.6k)	0	0	0	0	0	0	0/428	

01

TR1/TS0 TR0/TestSetB

02

blocklist

03

split GT SHA256 checkpoint split

04

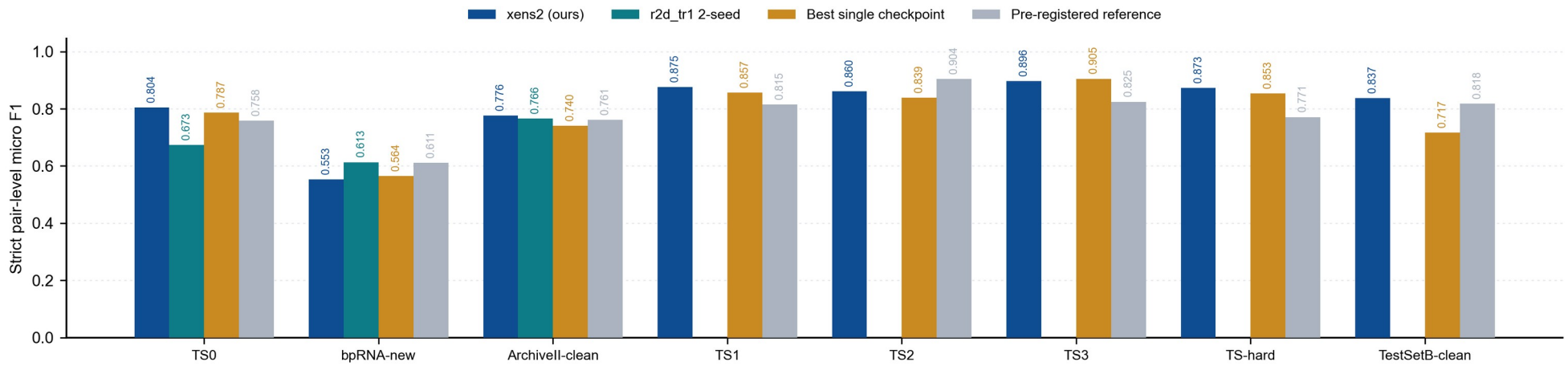
TestSetB clean-181 xens2 0.8370 vs RiNALMo-ft 0.8177

benchmark

1 xens2 5/8 checkpoint 4/8 bpRNA-new TS2 ArchiveII

split

a Eight-split benchmark: model identities are explicit



b The same values in tabular form (dash = not measured)

Split	xens2	r2d_tr1 2-seed	Best single ckpt	Clean reference
TS0	0.8039	0.6733	0.7866	RNAformer 0.7578
bpRNA-new	0.5532	0.6132	0.5643	UFold 0.6106
ArchiveII-clean	0.7760	0.7664	0.7403	RiNALMo-ft 0.7613
TS1	0.8755	—	0.8570	RNAformer-if 0.8150
TS2	0.8604	—	0.8393	RNAformer-if 0.9043
TS3	0.8965	—	0.9049	RNAformer-if 0.8245
TS-hard	0.8732	—	0.8530	RNAformer-if 0.7709
TestSetB-clean	0.8370	—	0.7167	RiNALMo-ft clean-181 0.8177

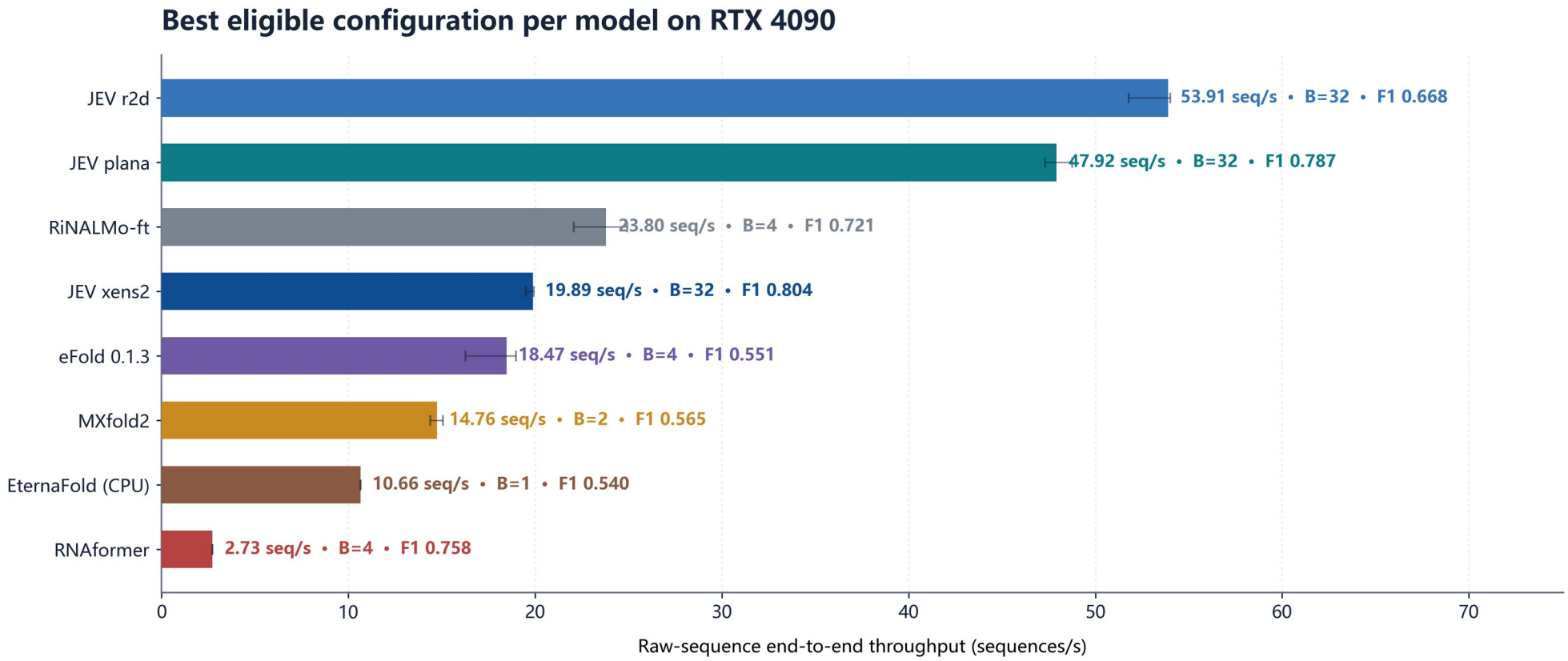
r2d_tr1: 2/3 above selected references but 0/3 on the full-clean sensitivity board; legacy TR1 is not a uniform clean-corpus model.

RNAformer original reports: TS0=0.728; TS1/TS2/TS3/TS-hard=0.739/0.802/0.702/0.662 (full-pair F1; not mixed into matched-GT SOTA counts).

new CONTRAfold 0.6968 EternaFold 0.6854 xens2 TS2 MXfold2 0.8991 ArchiveII Kirigami 0.7872 > xens2 0.7760 clean 6/8

2 RTX 4090

7 GPU + 1 CPU TS0 n=1,288 raw-sequence GPU



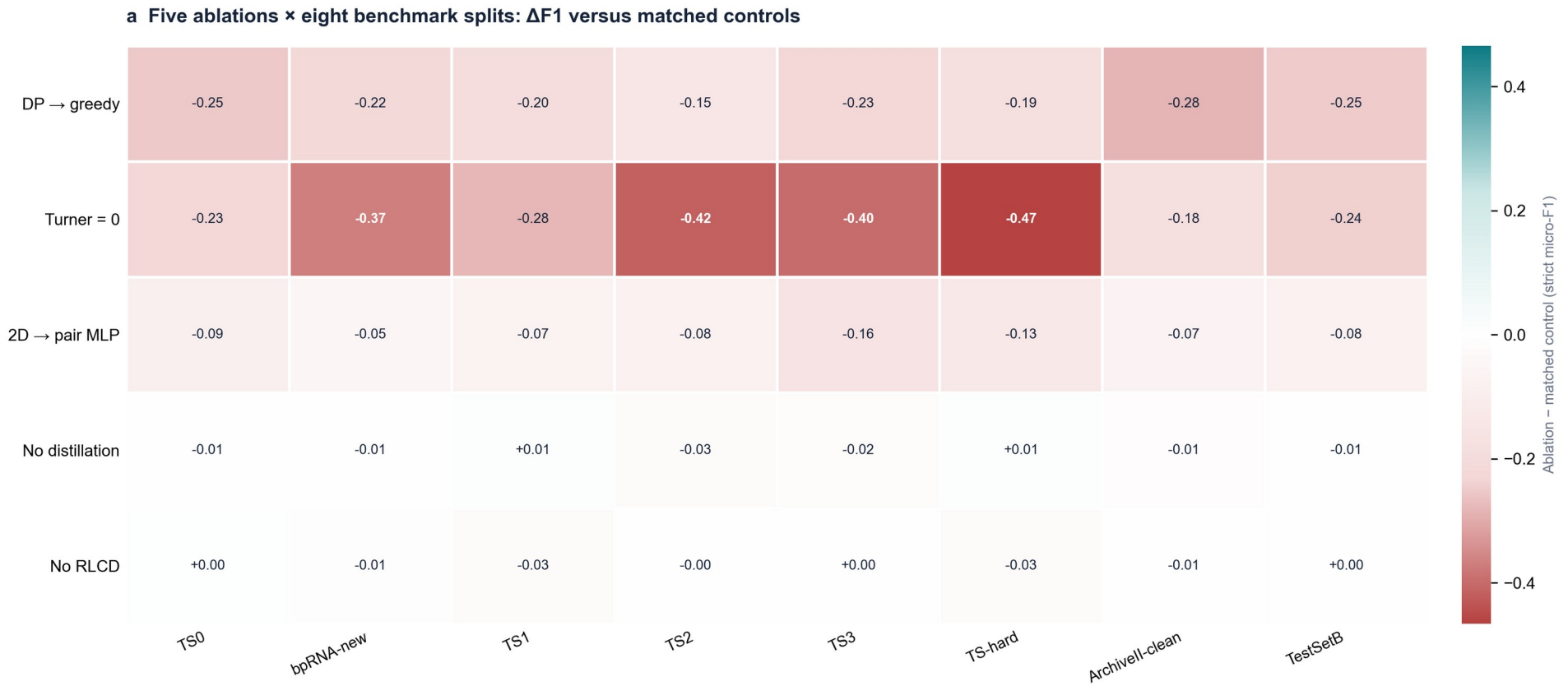
Full TS0 (n=1,288), three repeats; bars show repeat range. Raw sequence -> encoder -> task head -> required decode; no cached embeddings.
EternaFold uses its official serial CPU implementation (B=1). BPfold/UFold/E2Efold are blocked by unavailable official weights; no absolute SOTA claim.

r2d 53.91 seq/s eFold 0.1.3 B=4=18.47 EternaFold CPU=10.66 BPfold/UFold/E2Efold blocked SOTA

round38 JSON TS0 raw-sequence blocked

3a × 8 split

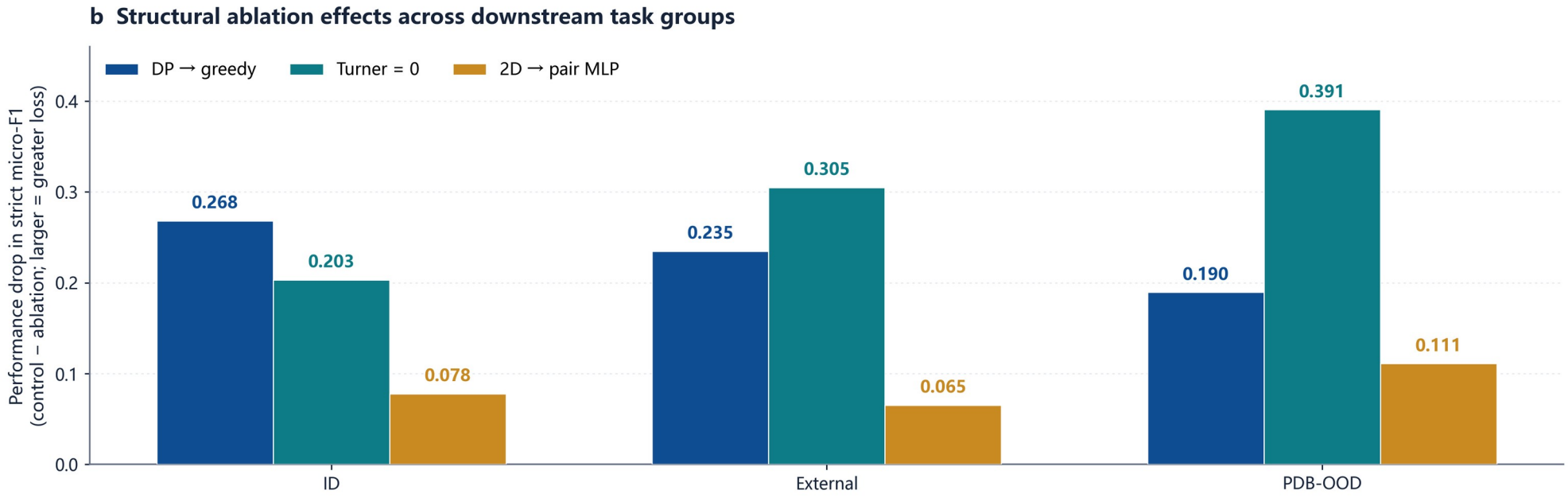
No RLCD B=2



No RLCD is compared only with its matched B=2 control; every cell is measured on the same RTX 4090 protocol.

– strict micro-F1 split

3b Turner/2D OOD DP



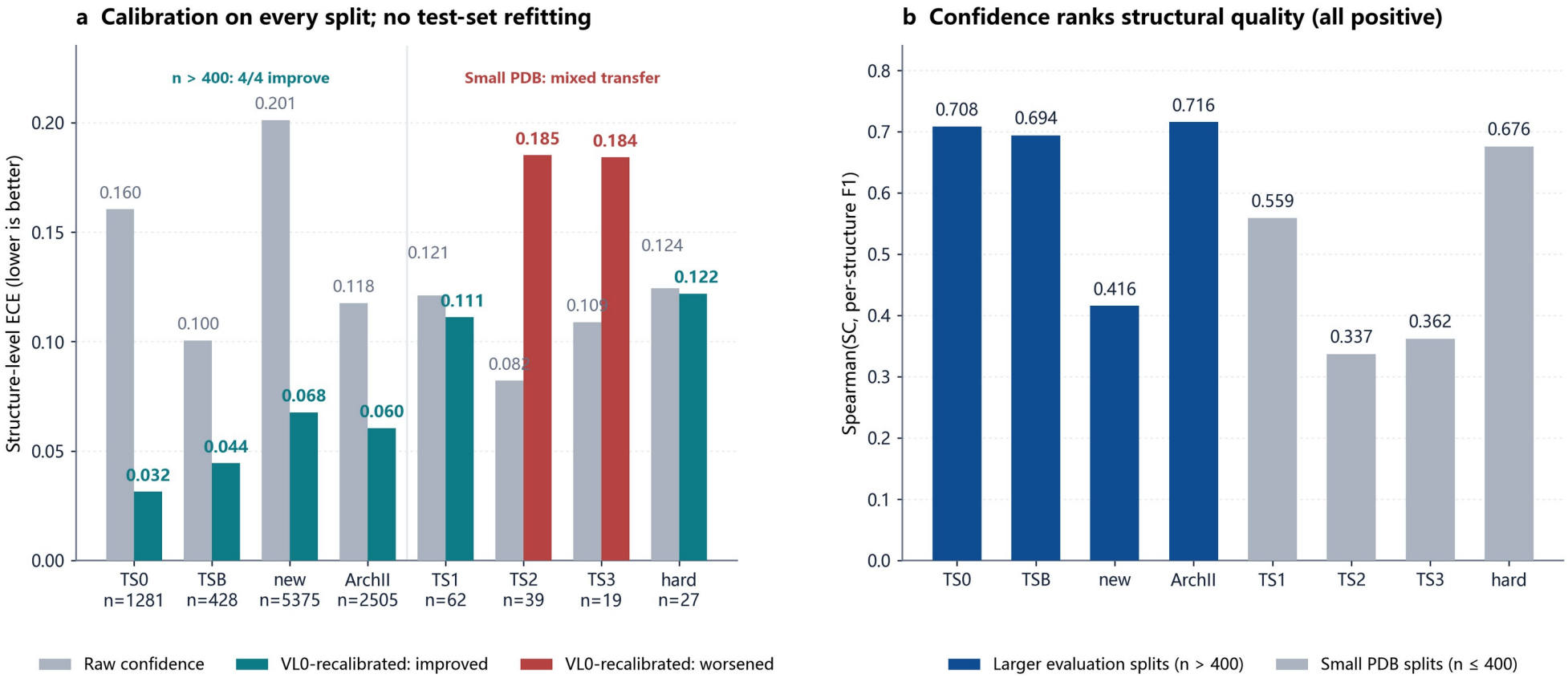
Task group	Included splits	DP drop	Turner drop	2D-scorer drop
ID	TS0 + ArchiveII-clean	0.268	0.203	0.078
External	bpRNA-new + TestSetB	0.235	0.305	0.065
PDB-OOD	TS1 + TS2 + TS3 + TS-hard	0.190	0.391	0.111

Turner 2D scorer PDB-OOD ID DP ID ID=TS0/ArchiveII =new/TestSetB PDB-OOD=TS1/2/3/hard

main_ablation_5x8.json PDB/ID

4 split PDB split

VL0 logit split ECE

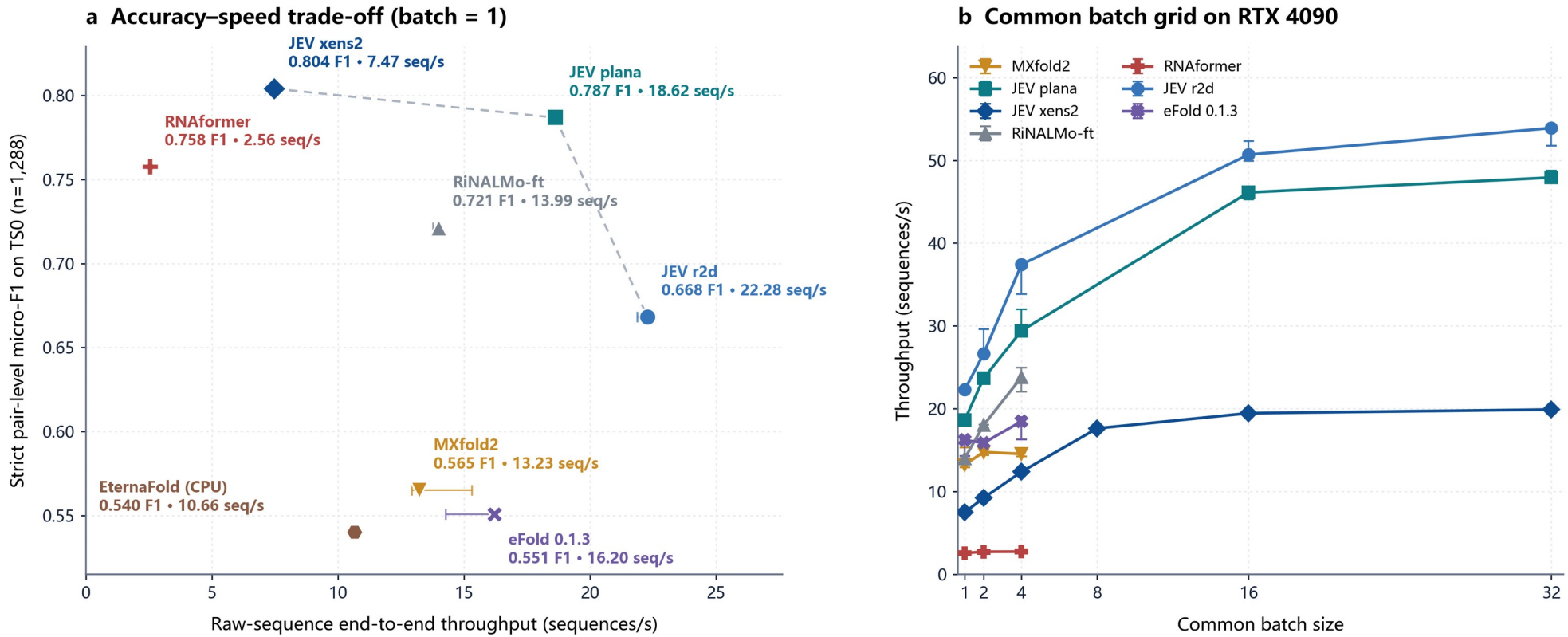


ECE uses 10 confidence bins. VL0 parameters a=1.6 and b=-1.9 are frozen across all eight test splits.

TS0/TestSetB/bpRNA-new/ArchiveII TS1 TS-hard TS2/TS3 n

5 RTX 4090 - batch

4090 TS0 n=1,288 strict micro-F1 raw-sequence a B=1 Pareto b batch

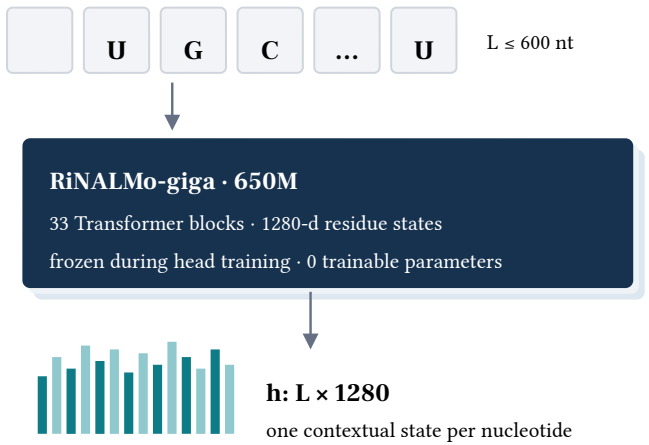


Only full-TS0 three-repeat results are plotted; cross-batch pair agreement >=99% and |delta F1| <=1e-3. Median and repeat range shown.
EternaFold is CPU-only and appears only in panel a. eFold is official 0.1.3; the old 0.1.2 board uses a different protocol and is checked separately.
BPfold/UFold/E2Efold remain blocked by official weights; no unrestricted speed SOTA claim. Legacy head-only values are excluded.

JEV eFold B=1/2/4=16.20/16.29/18.47 seq/s ≥ 99.97% EternaFold CPU=10.66 blocked SOTA

6

a FROZEN SEQUENCE REPRESENTATION



cached or recomputed on raw sequence

Input

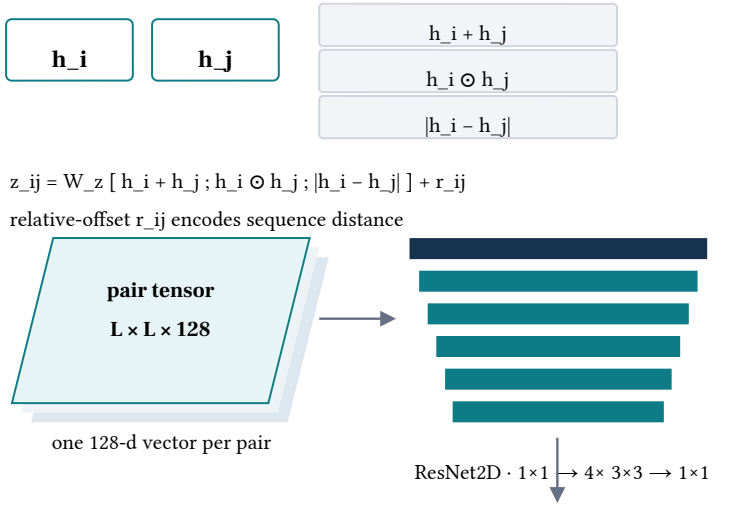
raw RNA sequence, canonical A/C/G/U

Output

contextual nucleotide states h

backbone never updated by downstream labels

b PAIRWISE 2D DECISION



z_ij = W_z [h_i + h_j ; h_i ⊙ h_j ; |h_i − h_j|] + r_ij

relative-offset r_ij encodes sequence distance

training objective

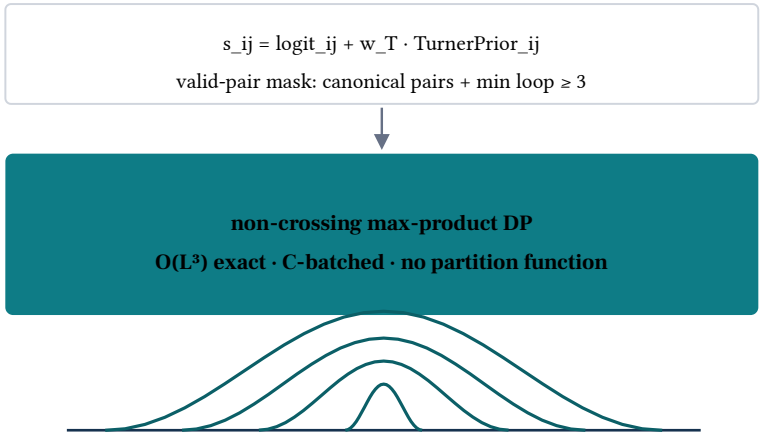
NLL + KL distillation + RLCD + soft-ECE

decision head · 812,375 trainable parameters

paired labels are symmetric; no pair-type loss is applied here

Output pairwise base-pair logits

c CONSTRAINED DECODING ✦ CALIBRATION



decoded structure · every arc is a legal base pair

STRUCTURE

dot-bracket

((..))

CONFIDENCE

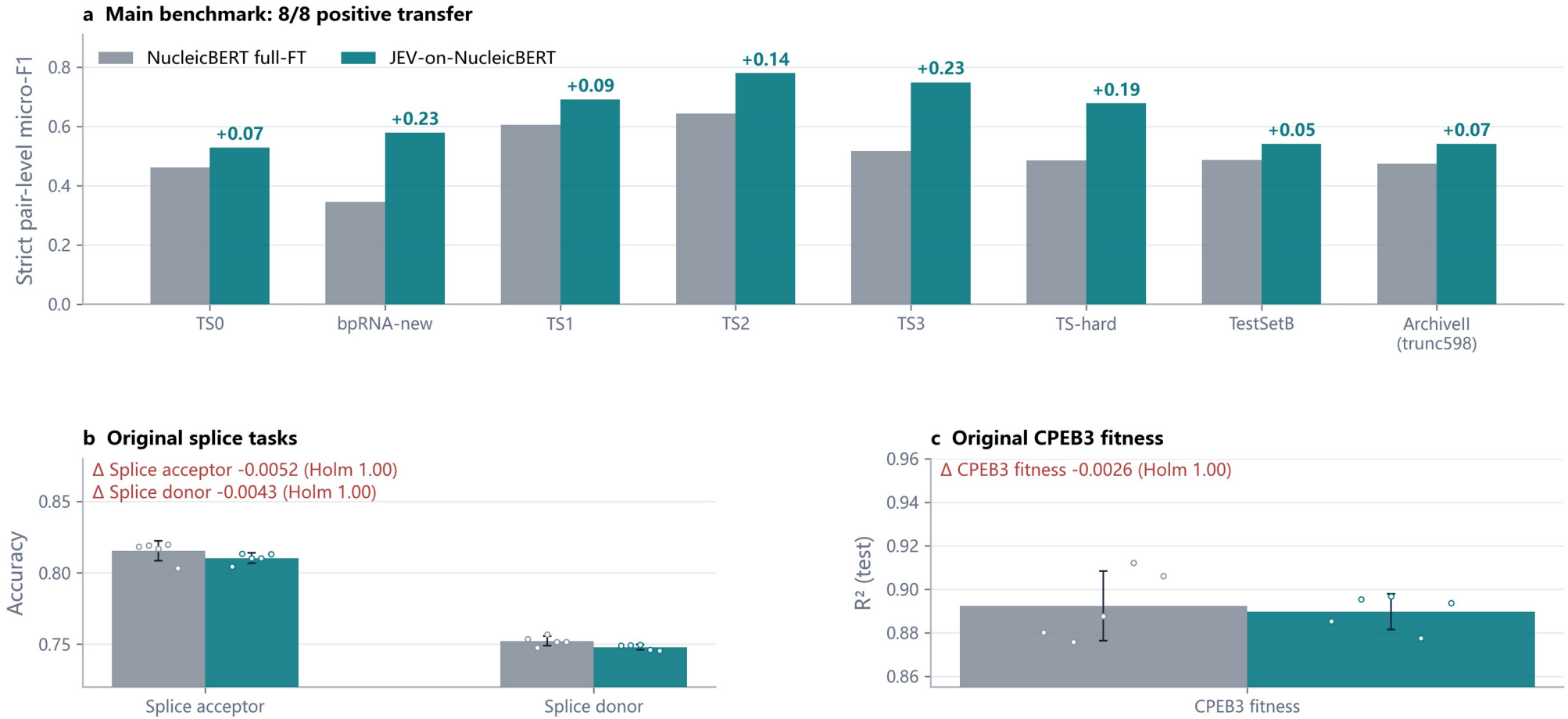
σ(logit) → VL0 affine → per-structure ECE

structure branch and calibration branch share one forward pass

legal by construction · no post-hoc repair

7 benchmark 8/8

benchmark acceptor/donor/CPEB3 contact/distance blocked

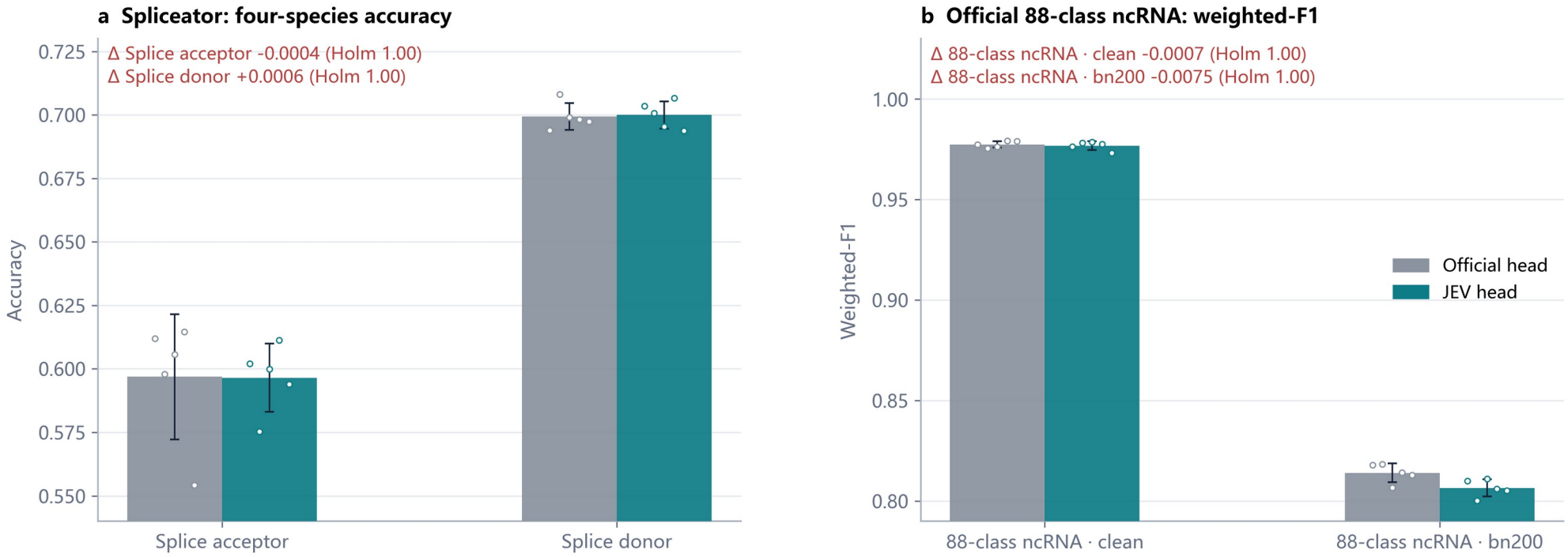


Five paired training seeds per head; bars are mean ± SD, dots are individual seeds. Holm correction within the original-task family. Main benchmark uses full-FT values.

8/8 benchmark secondary-structure contact/distance split blocked shuffle

8 Spliceator 88 ncRN

+ Holm acceptor −0.0004 donor +0.0006 clean/bn200 8B



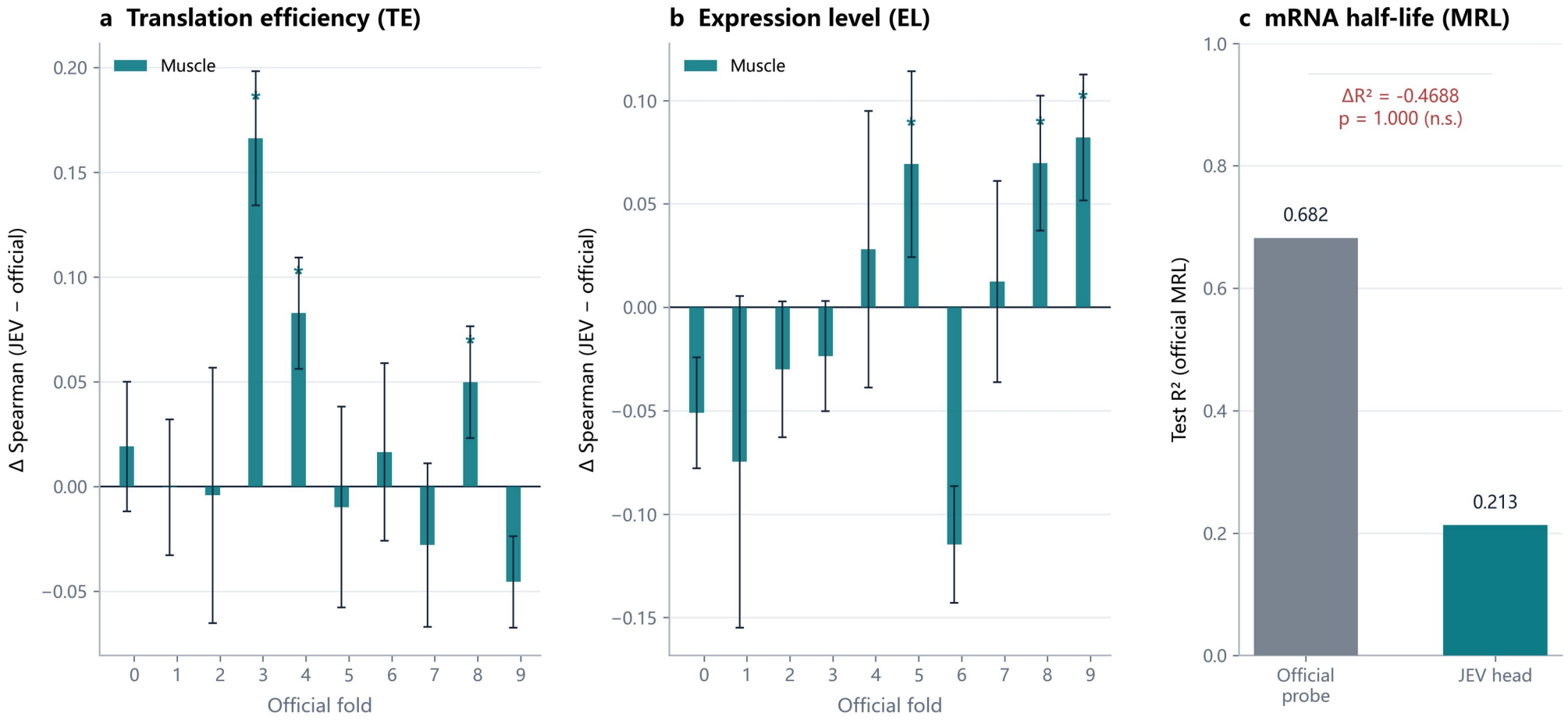
Full-dataset, five paired seeds per head; bars are mean ± SD, dots are individual seeds. Holm correction across the four classification tasks; all negative results are retained. ArchiveII family-level structure and the 8B regression panel are reported separately; a non-significant positive donor delta is not a positive claim.

ArchiveII tRNA/5s T-A148 OOD pair-density bpRNA TS0 8B Muscle

round33/34 full results + external QA 8A 8B

8B Muscle TE ✦0.025 EL/MRL

Muscle TE +0.0246 5/10 Holm 0.86 EL −0.0033 5/10 MRL −0.4688 HEK/pc3

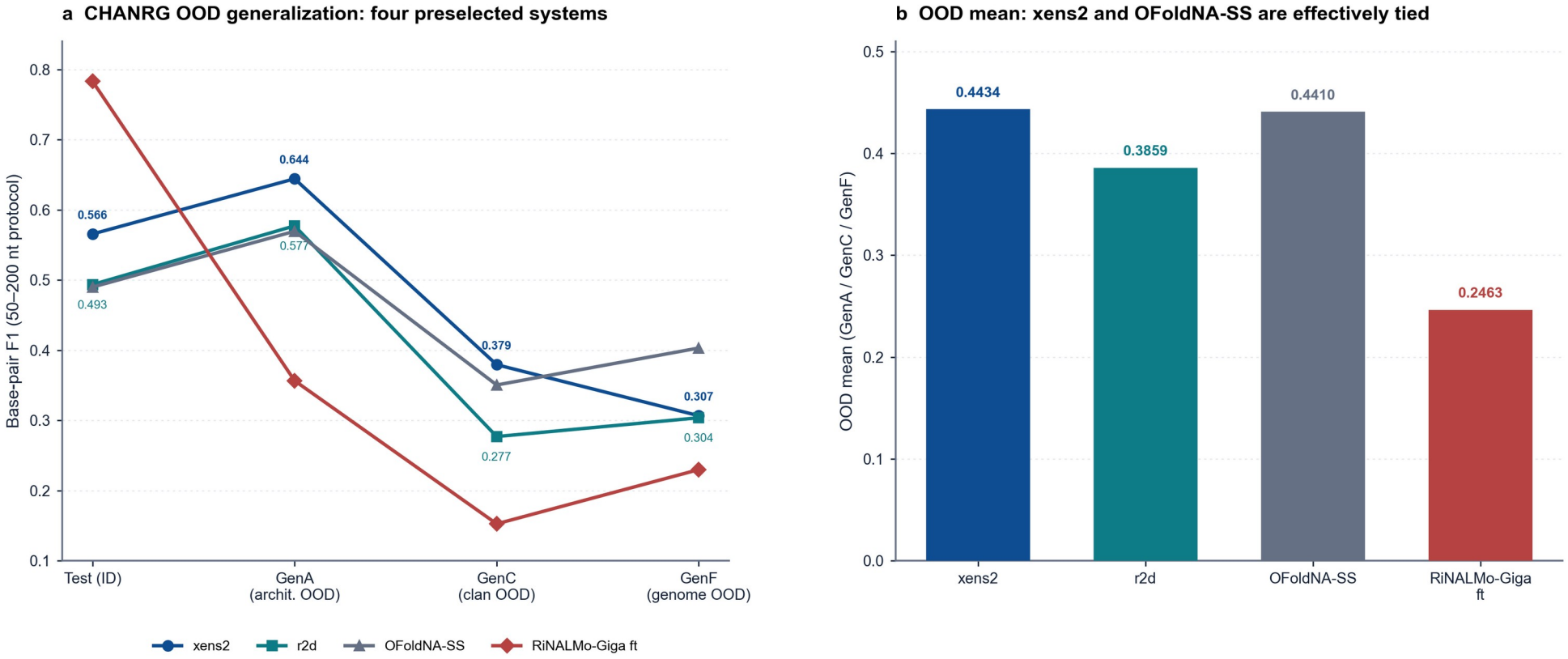


HEK/pc3 fold3 EL pc3 fold1 8A

All ten folds shown; error bars = five-seed SD; * one-sided p<0.05; Holm is descriptive; HEK/pc3 not plotted

9 CH NRG OOD mean GenF

a 26 b OOD mean

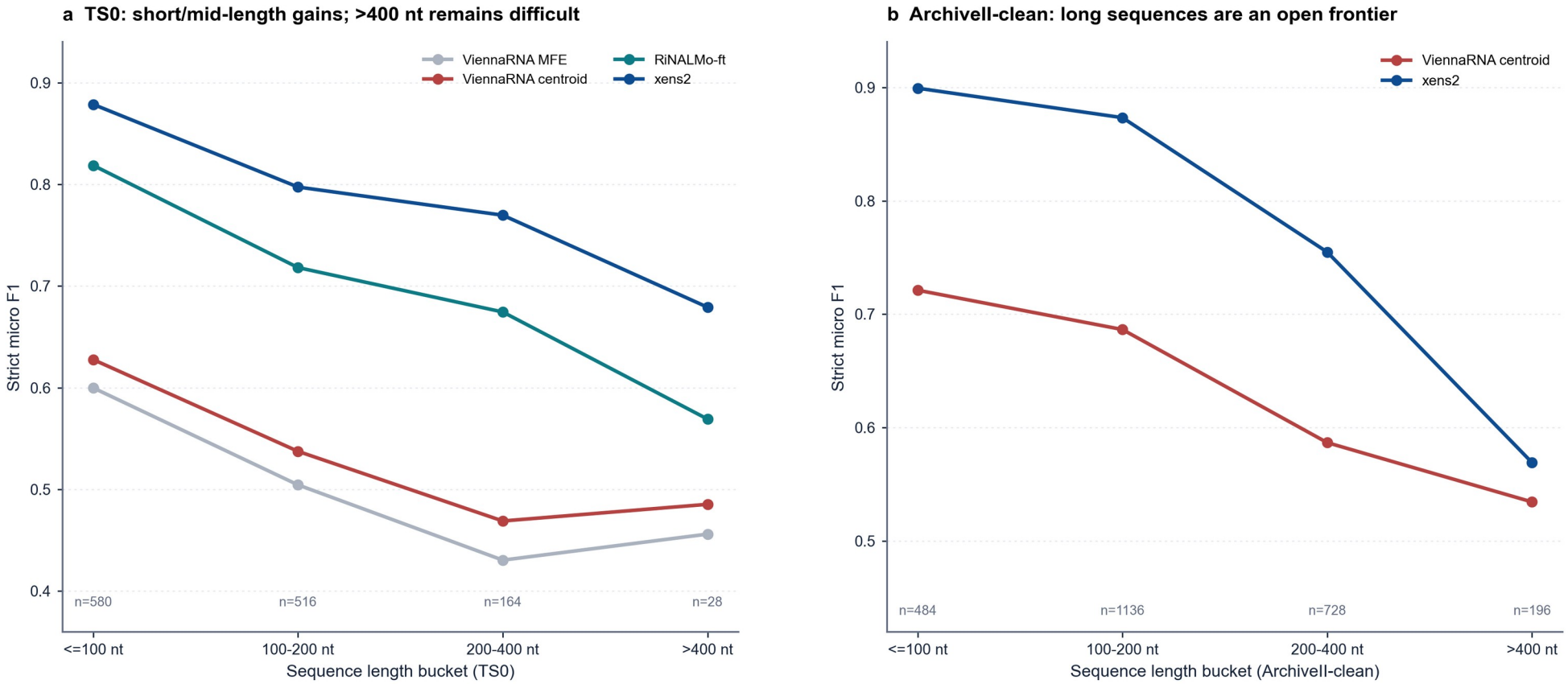


xens2 OOD mean = 0.4434 versus OFoldNA-SS = 0.4410 (+0.0024): a tie, not a superiority claim. GenF remains a weakness.

GenA=0.6443 GenC=0.3793 GenF=0.3066 OOD mean=0.4434 20.8%/39.2%

10 ◂ 400 nt

TS0 ArchiveII-clean strict micro-F1



Retention is descriptive, not a claim of long-sequence superiority. Bucket counts are printed beneath each panel.

retention	35

claim–evidence–limitation

创新线	可支□的结论	必须□时展示的限制
benchmark	xens2 5/8 checkpoint 4/8	clean 6/8 bpRNA-new TS2 ArchiveII
	4090 raw-seq fp32 B=32 53.91/47.92/19.89 seq/s JEV -	RNAformer B=1/2/4=2.56/2.72/2.73 UFold SOTA batch
OOD	Turner/2D PDB-OOD DP CHANRG	GenF TS2 xens2 OFoldNA-SS OOD mean
ECE	split	TS2/TS3
	NB benchmark 8/8 acceptor/donor/CPEB3 8B Muscle TE fold3/4 EL fold5	NB contact/distance split blocked shuffle MRL Muscle TE/EL HEK fold3 pc3

→ OOD ↠

xens2=5/8 checkpoint=4/8

xens2 r2d_tr1

	TS0	new	TS1	TS2	TS3	hard	rchII	TSB	
ViennaRNA mfe	0.5056	0.6379	0.7209	0.8980	0.8206	0.7663	0.5752	0.5225	CPU
ViennaRNA centroid	0.5393	0.6770	0.7299	0.8930	0.8235	0.7958	0.6181	0.5482	CPU
ViennaRNA mea	0.5241	0.6620	0.7192	0.8938	0.8237	0.7864	0.6054	0.5393	CPU
Nussinov+Turner	0.2126	0.3015	0.3730	0.4187	0.4131	0.4067	0.2000	0.2164	CPU
MXfold2	0.5651	0.6688	0.7967	0.8991	0.8127	0.8028	0.7400	0.6504	GPU
EternaFold	0.5400	0.6854	0.7426	0.8635	0.8005	0.7658	0.6157	0.5850	CPU
UFold	0.6598	0.6106	0.6455	0.8792	0.6326	0.6388	0.4111	0.5858	GPU
RNAformer (bprna)	0.7578	0.4936	0.7658	0.8590	0.9410‡	0.7845‡	—	—	GPU
RNAformer-if GT	—	—	0.8150	0.9043	0.8245	0.7709	—	—	GPU
RNAformer-biophys GT	—	—	—	—	0.8265	—	—	—	GPU
RiNALMo-ft (Zenodo)	0.7214	0.4489	0.7139	0.6702	0.6055	0.5556	0.7613	0.8177†	GPU
NucleicBERT frozen+	0.1509	0.0955	0.2029	0.1541	0.1439	0.1343	0.2372	0.1210	GPU
NucleicBERT fine-tuned	0.4609	0.3453	0.6049	0.6435	0.5175	0.4845	0.4735	0.4862	GPU
eFold (Sci Adv 26)	0.5612	0.6640	0.7682	0.8862	0.7354	0.7632	0.6618	0.5661†	GPU
DEPfold (ICLR 25)	0.4430	0.1736	0.3735	0.3270	0.2670	0.2409	0.6216	0.5835†	GPU
RFold (ICML 24, TR0)	0.5520	0.5313	0.6578	0.7002	0.6495	0.6220	0.6811	0.5740†	GPU
Kirigami (arXiv 2406)	0.7386	0.5800	0.7366	0.7061	0.7436	0.6782	0.7872	0.7514†	GPU
xens2	0.8039	0.5532	0.8755	0.8604	0.8965	0.8732	0.7760	0.8370†	GPU
r2d_tr1 2-seed	0.6733	0.6132	—	—	—	—	0.7664	—	GPU

RNAformer-if/biophys checkpoint canonical nested GT Table 4 † =clean-181 ‡ = clean

inter-fam/biophys checkpoint GT Table 4 1

clean

19 × 8 split

	TS0	new	TS1	TS2	TS3	hard	ArchII	TSB	
ViennaRNA mfe	0.5056	0.6379	0.7209	0.8980	0.8206	0.7663	0.5752	0.5225	
ViennaRNA centroid	0.5393	0.6770	0.7299	0.8930	0.8235	0.7958	0.6181	0.5482	
ViennaRNA mea	0.5241	0.6620	0.7192	0.8938	0.8237	0.7864	0.6054	0.5393	
Nussinov+Turner	0.2126	0.3015	0.3730	0.4187	0.4131	0.4067	0.2000	0.2164	
MXfold2	0.5651	0.6688	0.7967	0.8991	0.8127	0.8028	0.7400	0.6269†	○
EternaFold	0.5400	**0.6854**	0.7426	0.8635	0.8005	0.7658	0.6157	—	○
UFold	0.6598	0.6106	0.6455	0.8792	0.6326	0.6388	0.4111	0.5331†	○
RNAformer (bprna)	0.7578	0.4936	0.7830†	0.8506†	0.8296†	0.6933†	—	×	
RNAformer-if GT	—	—	0.8150‡	0.9043‡	0.8245‡	0.7709‡	—	—	○
RNAformer-biophys GT	—	—	—	—	0.8265‡	—	—	—	○
RiNALMo-ft (Zenodo)	0.7214	0.4489	0.7139	0.6702	0.6055	0.5556	0.7613	0.8177†	○
NucleicBERT frozen+	0.1509	0.0955	0.2029	0.1541	0.1439	0.1343	0.2372	0.1210	○
NucleicBERT fine-tuned	0.4609	0.3453	0.6049	0.6435	0.5175	0.4845	0.4735	—	○
eFold 0.1.2 (Sci Adv 26)	0.5612	0.6640	0.7682	0.8862	0.7354	0.7632	0.6618‡	0.5661†	○
DEPfold (ICLR 25)	0.4430	0.1736	0.3735	0.3270	0.2670	0.2409	0.6216	0.5835†	○
RFold (ICML 24, TR0)	0.5520	0.5313	0.6578	0.7002	0.6495	0.6220	0.6811	0.5740†	○
Kirigami (arXiv 2406)	0.7386	0.5800	0.7366	0.7061	0.7436	0.6782	0.7872	0.7514†	○
xens2	**0.8039**	0.5532	**0.8755**	0.8604	**0.8965**	**0.8732**	**0.7760**	**0.8370†**	○
r2d_tr1 2-seed	×	0.6132	—	—	—	—	×	—	○

†=clean RNAformer TS1/TS2/TS3/hard n=25/11/4/8 TSB n=181 ×= — = / ○ = ‡ = checkpoint/

B / head-only

124.78 22.28 4090

head-only	seq/s	F1	
B=1	21.45	0.667971	/
B=32 + C	96.14	0.667971	/
B=64 fp16	119.55	0.667971	/
B=128	121.24	0.667971	/
B=192	120.86	0.667971	/
B=256	124.78	0.667971	
B=384	116.40	0.667971	
B=512	122.60	0.667971	

head-only

cached embedding → 812,375 head → C 4090 117.735 seq/s

forward-only

raw sequence → + → / DP RiNALMo-ft 14.5 seq/s

raw sequence

raw sequence → + + DP r2d B=1 22.28 seq/s

B=1 21.45 vs 22.28 4090 117.735 seq/s

JEV 4090 r2d B=32 53.91 seq/s eFold/EternaFold blocked SOTA

Jev

		TS0 F1	
r2d / tr0	frozen RiNALMo + ResNet2D	0.6629	
r2d / tr1c	split	0.6679	
plana / tr1c		0.7866	ID OOD
xens2	plana × r2d w=0.7	0.8039	
Round 32		6/8 clean	7/8

J2 ·

L×L pair logits

J3 ·

DP 0

J4 ·

VL0 + SC/ECE

J6 · System-1/2

System-1 /

J1 ·

+DP

DP OOD

matched_30k mixed

arm	Test	Gen	GenC	GenF	OOD mean	
clean_s0	0.6395	0.4772	0.2247	0.2582	0.3200	3-seed
mixed_s0	0.5809	0.5671	0.2820	0.2325	0.3605	3-seed
clean_s1	0.6274	0.4604	0.2674	0.2771	0.3350	3-seed
mixed_s1	0.5854	0.5755	0.2855	0.2584	0.3732	3-seed
clean_s2	0.6250	0.5273	0.2627	0.2664	0.3521	3-seed
mixed_s2	0.5597	0.5396	0.2392	0.2835	0.3541	3-seed

T-A127 · matched

24 3-seed mean±std Wilcoxon–Holm

T-A128 · 40/40

40/40 B4/B2 8/8 ID/ /PDB-OOD

T-A129 · RiNALMo

MRL HEK fold0–3 TE/EL QA HEK 4 pc3

·

GitHub round29-handoff round35 24 PASS SHA256

splice GenF

01
tr1c
split GT SHA256
02
benchmark
xens2 5/8 4/8
bpRNA-new TS2 ArchiveII
r2d_tr1
6/8 7/8
03
raw-seq 53.91/47.92/19.89
seq/s
JEV
RNAformer
B=1/2/4 eFold/EternaFold
blocked SOTA 124.78
head-only
04
5×8
Turner/2D OOD
DP
distillation/RLCD
05
ECE NB
benchmark 8/8
22
CHANRG OOD mean
GenC/GenF
20.8%/39.2% >400 nt

把每个数字放回它所属的

模型、数据集和协议

01 模型身份 xens2 5/8 checkpoint 4/8 bpRNA-new TS2 ArchiveII

02 科学证据 OOD

23 可复现交付 result.json / GitHub

10 · 5 · 1 · · 1.5

END